

Experiment :-

Study of LAN Topology Network devices and Address configuration.

Objective :-

- (1) To study the LAN topology used in college campus.
- (2) To identify networking device & understand their functions.
- (3) To analyze wired and wireless communication method used in the network.
- (4) To measure speed of network for wired and wireless connections.
- (5) To obtain MAC, IP Address & subnet mask.

Problem statement - I

Draw & Identify LAN topology of college campus and clearly label all components such as router, switcher & etc.

Soln:- The college campus network observed this study consists of :-

- (i) Layer-3 core switcher in the server room.
- (ii) Managed by PoE+ switcher.
- (iii) Wireless Access Point (APs)
- (iv) Ethernet port in classroom and laboratories.

Network topology :-

- (i) Core layer
- (ii) Distribution layer
- (iii) Access layer

These all are the network hierarchical layer.

Hybrid Hierarchical Topology which combines,

- (i) Tree topology
- (ii) Extended Star topology.
- (iii) Ring Topology (Server room)

Core layer :-

Core layer contains server room and server room have firewall, router, 2-34 switches patch Rack, servers, printer etc.

Distribution layer :-

Distribution layer contains server Room and the 4 main switches are -

- (i) Main Build → R7 (Room no - 213, 2nd Floor)
- (ii) " → R1 (" 301, 3rd ")
- (iii) " → R6 (" 113, 1st ")
- (iv) " → R2 (" 201, 2nd ")

→ This 4-Switches contains

(i) 18 Port Gigabit Easy smart switch with 16-port

[TL-SG1218-P1]

(ii) [SC3428xMP] → 24 Port Gigabit - 8 4-port 10Gb SFP+, (2+ Managed switcher with 24 port PoE)

→ Now lets discuss these 2 above components →

Now IP Links TL-SR1218 MP8

(i) Linker used for :- CCTV camera, wifi access point etc.

(ii) 18 port Gigabit "Easy Smart" switch

(iii) 16 port support PoE +

TR - Link Omada SG3428xMP

(i) Higher-end managed Layer 2+ switch

(ii) 24 PoE + Port, Unlinks, 10G SFP

(iii) Usually acts as a main distribution for each floor.

Access Layer :-

Access layer contain 3- Switcher i.e.

(i) Main Build R1 - [Room no - 301, 3rd floor]

(ii) " R4 - [Room no - Ground floor]

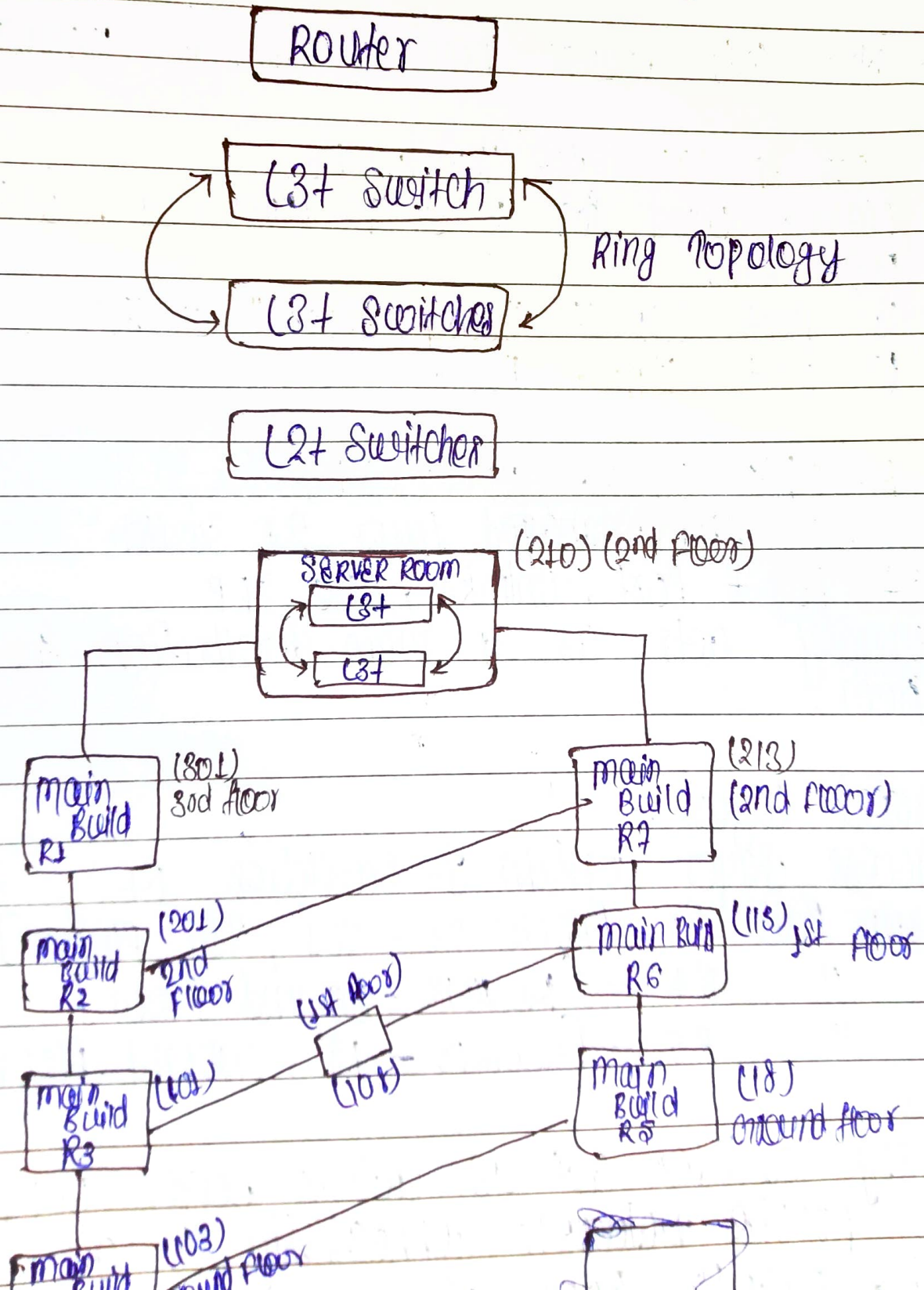
(iii) " R5 - [Room no - 18, Ground floor]

• Every switcher act as access layer which helps to provide network connectivity to all class room lab, faculty Room etc

Access layer directly connect :-

- (i) Ethernet port
- (ii) Wireless Ap
- (iii) Lab System
- (iv) Outdoor wireless AP (or - OLS)

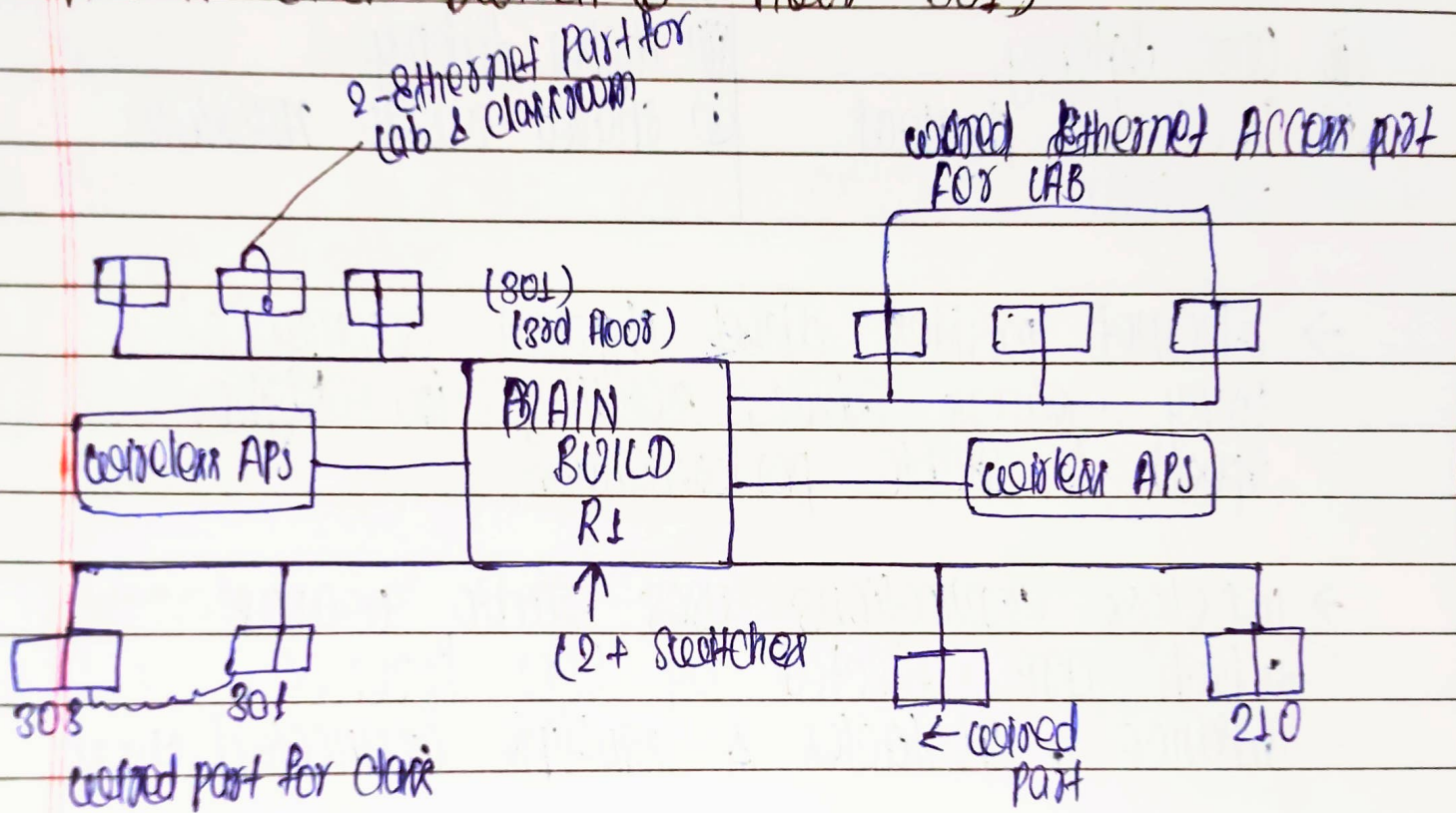
→ Server Room Architectural Topology :-





- R1, R7, R2, R6 → Distribution layer.
- Server Room → Core layer
- R1, R2, R3, R4, R5, R6, R7 → Access layer

ACCESS LAYER SWITCH (3rd Floor - 301)



[illegible]



```
C:\Users\Hp>netsh wlan show interface
```

```
There is 1 interface on the system:
```

```
Name                : Wi-Fi
Description          : Realtek RTL8822CE 802.11ac PCIe Adapter
GUID                 : d3b9d51c-4c0b-48d4-ae5e-6f631e7e8b17
Physical address     : e8:fb:1c:57:8e:e3
Interface type       : Primary
State                : connected
SSID                 : Haweli
AP BSSID             : 72:ca:e7:a2:d3:30
Band                 : 5 GHz
Channel              : 44
Connected Akm-cipher : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11ac
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode       : Auto Connect
Receive rate (Mbps)  : 54
Transmit rate (Mbps) : 54
Signal               : 100%
Rssi                 : -34
Profile              : Haweli
QoS MSCS Configured  : 0
QoS Map Configured    : 0
QoS Map Allowed by Policy : 0
```

Difference b/w wired & wireless Speed?

wired Network

wireless Network.

- (i) Faster Communication
- (ii) Stable Connection
- (iii) Low latency
- (iv) Dedicated physical medium

- (i) slower Communication
- (ii) Affected by distance/Interference.
- (iii) Higher latency
- (iv) shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q. Role of subnet mask in network?

A subnet identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

• It is important for :-

- (i) Network segment
- (ii) Efficient routing
- (iii) Communication b/w devices.

OBSERVATION

During the study of the college network infrastructure :-

- i) A centralized server room was used of network backbone.
- ii) Two LG switches were connected in redundant cyclic structure.
- iii) managed IP-Link PDE + switches were used " network Cabinets.

Advantage of Existing Network?

- i) Scalable Architecture
- ii) Centralized management
- iii) Support PDE + device
- iv) Easy expansion of floors & classroom
- v) Support both wireless & wired.
- vi) Redundant link for failover.

Conclusion.

The observed campus can follow a Hybrid hierarchical topology consist of tree, extended star, Ring, the network uses manage of PDE switches wireless access - point ethernet port & centralized server infrastructure to provide efficient wired & wireless communication, access the campus.